Bharath Mandha

AI/ML Engineer, 4+ years building production ML and LLM systems
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EXPERIENCE

Goldman Sachs, AI/ML Engineer — Jun 2025 to Present

- Designed the retrieval and LangGraph orchestration layers and their FastAPI integration for a RAG platform over 3M+ internal financial documents (tens of millions of chunks), used weekly by 70+ credit and risk analysts
- Cut average credit-memo preparation time by about 40% by automating evidence discovery, document analysis and validation, with analyst review preserved
- Built a 500+ query RAG evaluation benchmark (factual retrieval, financial comparisons, evidence-based reasoning) scoring retrieval with Hit Rate and MRR and generation with RAGAS faithfulness, groundedness and answer relevance, apart from the 10 to 20 s end-to-end workflow; used it to compare chunk size, overlap, top-k and embedding choices experiment by experiment, alongside metadata-filtering and index-performance work; groundedness above 90%
- Built LangGraph agents for retrieval, analysis, validation and delegation that took over repetitive credit-analysis reasoning steps, with analysts reviewing sensitive decisions
- Built components of a PySpark and AWS S3 ingestion pipeline (preprocessing, recursive and semantic chunking, metadata enrichment, embedding generation, incremental vector-index updates) so new or modified documents reached the index without a full rebuild for routine updates
- Evaluated quantization and LoRA/QLoRA approaches for open-source models to assess GPU-memory requirements, inference latency, model quality and serving cost, and concluded that infrastructure savings alone didn't justify moving complex credit reasoning to smaller models

Cigna, Machine Learning Engineer — Mar 2022 to Aug 2023

- Built 30-day hospital readmission-risk prediction with XGBoost and LightGBM over 1.5M+ HIPAA-regulated encounters, raising ROC-AUC from 0.85 to 0.91 on a held-out cohort of 200K+ encounters from 70K+ patients with time-aware, patient-level splits
- In a fixed-capacity planning scenario, identified about 15% more high-need patients than the previous prioritization approach through feature engineering, class-imbalance handling and precision-recall threshold optimization
- Engineered 150+ candidate features from labs, demographics, encounter history and clinical notes; about 100 kept in production for predictive value and stability
- Productionized training, batch inference, registration and retraining (MLflow, Airflow) on SageMaker and EKS through CI/CD with PSI drift monitoring; SHAP explanations let clinical reviewers prioritize several thousand high-risk patients monthly

Best Buy, Associate Data Scientist — Jun 2020 to Feb 2022

- Cut weighted MAPE of SKU-and store-level demand forecasts by about 14% relative with gradient-boosted models benchmarked against time-series baselines on time-based holdouts, tracking RMSE alongside MAPE
- Built PySpark, SQL and BigQuery pipelines over 50M+ transactions across thousands of SKUs and hundreds of stores
- Raised response on comparable campaigns by about 12% with six customer segments from clustering and cohort analysis

PROJECTS (CODE, RESULTS AND CI PUBLIC ON GITHUB)

finance-rag-eval, retrieval over SEC filings — 2026

- Page-level retrieval evaluation on FinanceBench (150 questions, 360 filings, 53,399 pages) with a held-out split and paired bootstrap intervals: a company and period filter (+0.29) and statement-page routing (+0.28) take Hit@10 from 0.14 to 0.75
- A MiniLM hybrid led on dev but tied BM25 on test (0.73 vs 0.75) at about 5x the latency, so the service runs BM25: 126 queries/s per core at 12 ms p95, about $0.10 per million queries; a browser port matches its scores bit for bit

readmission-capacity, outreach planning on public hospital data — 2026

- Readmission models judged at a fixed call capacity on 101,766 UCI encounters split by patient: a 10% LightGBM call list holds 25.0% of 30-day readmissions (95% CI 23.2 to 26.7%) vs 20.5% for prior stays and 15.8% for LACE, with 539 readmissions expected on the list and 569 observed
- TreeSHAP reason codes with code descriptions checked in CI against the dataset's code table; simulated shifts caught by PSI and missing-value alerts; CI re-downloads the data, retrains and fails if headline metrics move

SKILLS

Languages and data: Python, SQL, PySpark, BigQuery

ML and LLMs: PyTorch, scikit-learn, XGBoost, LightGBM, SHAP, Hugging Face, LoRA/QLoRA, LangGraph, LangChain, FAISS, RAGAS

MLOps and cloud: MLflow, Airflow, FastAPI, Docker, Kubernetes, Terraform, CI/CD, AWS (SageMaker, S3, EKS), GCP Vertex AI

EDUCATION AND CERTIFICATIONS

University of North Texas, Master of Science in Computer Science — Aug 2023 to May 2025

AWS Certified Machine Learning Engineer, Associate
Google Cloud Professional Machine Learning Engineer
Databricks Certified Generative AI Engineer Associate